

Technical specification

Valve-regulated lead-acid battery with Absorbent Glass Mat (AGM) separator. Sealed construction for standby power applications.

12 V

6 cells

33 Ah

10 h to 10.8 V at 25°C

AGM

Valve-regulated lead-acid

Electrical performance

Rated capacity	33 Ah at 10 h to 10.8 V, 25°C
3-hour capacity	24 Ah to 10.8 V at 25°C
1-hour capacity	19 Ah to 10.5 V at 25°C
Internal resistance	11 mohm at 25°C
Maximum discharge	330A for 5 seconds at 25°C
Self-discharge	<=3% per month at 25°C
Design life / standby	10 years (JIS, 25°C); 10-12 years (Eurobat, 20°C)

Charging and operating conditions

Charge method	Constant voltage
Float / cycle voltage	13.5V-13.8V / 14.5V-15.0V at 25°C
Maximum charge current	9.9A
Temperature compensation	Float: -18 mV/°C per battery; cycle: -30 mV/°C per battery
Operating temperatures	Discharge: -15 to +50°C; charge: -10 to +50°C
Storage temperature	-20 to +50°C
Nominal operating temperature	25 ±3°C

Mechanical specification

Dimensions (L × W × H)	195±2 × 130±2 × 155±2 mm; overall height 167±2 mm
Approximate weight	10.00 kg ±4%
Terminal / torque	T6/T14; 5.1±0.6N.m
Container	ABS, UL94-HB; V-0 optional

Construction & applications

AGM construction with high-performance plates and electrolyte for standby and general-purpose power. The Yellow HR/HRL range uses reinforced grids and a special electrolyte formulation for UPS and intensive high-current discharge.

Battery construction

Positive plate	Lead dioxide
Negative plate	Lead
Container / lid	ABS / ABS
Safety valve	Rubber
Terminals	Copper
Separator	Glass fibre (AGM)
Electrolyte	Sulfuric acid

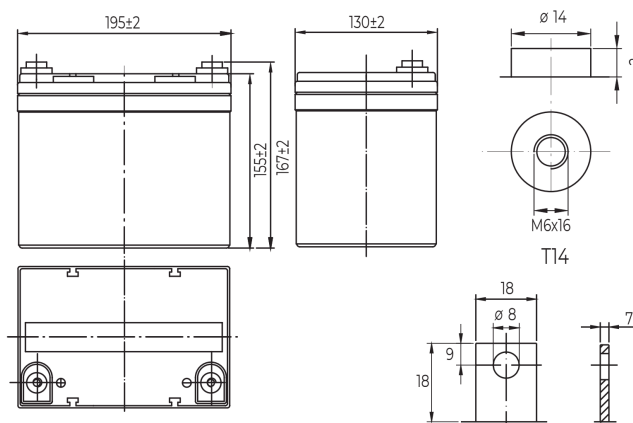
Features

- Insulated intercell connections.
- Gas recombination efficiency up to 99%.
- High-rate discharge performance.
- Special welding process.
- Self-regulating valves; no water topping-up.
- Shelf life without recharge: 6 months at 25°C.

Applications

- Uninterruptible power supplies (UPS).
- Telecommunications backup power.
- Power and energy infrastructure.
- Security and fire alarm systems.
- Renewable-energy systems.
- Medical equipment.
- Emergency lighting.
- Data centres.

Dimensions and terminal details / mm



Dimensions and tolerances govern; do not scale the drawing.

Discharge performance

Fully charged battery / 25°C

Constant-current discharge / A

V/cell	10m	15m	30m	1h	2h	3h	4h	5h	8h	10h	20h
1.60	71.90	55.40	33.20	20.40	12.10	8.68	6.93	5.92	4.07	3.36	1.78
1.65	69.80	54.10	32.50	20.10	12.00	8.63	6.89	5.89	4.05	3.35	1.78
1.70	66.90	52.10	31.50	19.60	11.90	8.57	6.85	5.85	4.02	3.34	1.77
1.75	64.00	50.30	30.70	19.00	11.70	8.51	6.80	5.81	3.99	3.32	1.76
1.80	60.40	47.70	29.60	18.40	11.40	8.26	6.59	5.63	3.87	3.30	1.75

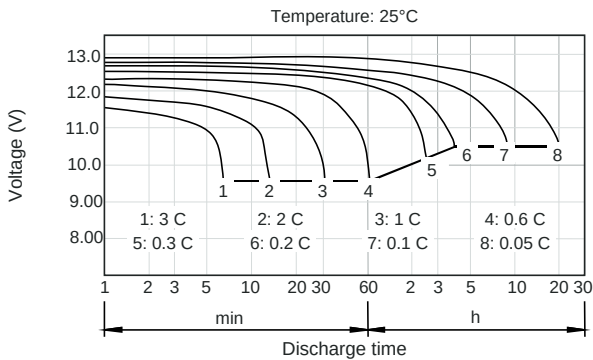
Constant-power discharge / W per cell

V/cell	10m	15m	30m	1h	2h	3h	4h	5h	8h	10h	20h
1.60	129.50	101.50	62.00	38.67	23.33	17.00	13.58	11.67	8.07	6.68	3.57
1.65	125.67	99.00	60.83	38.17	23.17	17.00	13.52	11.60	8.02	6.67	3.55
1.70	120.50	95.33	59.00	37.17	23.00	16.83	13.42	11.52	7.97	6.65	3.55
1.75	115.17	92.17	57.50	36.00	22.67	16.67	13.32	11.45	7.90	6.62	3.52
1.80	108.83	87.33	55.33	34.83	22.00	16.18	12.92	11.10	7.67	6.57	3.50

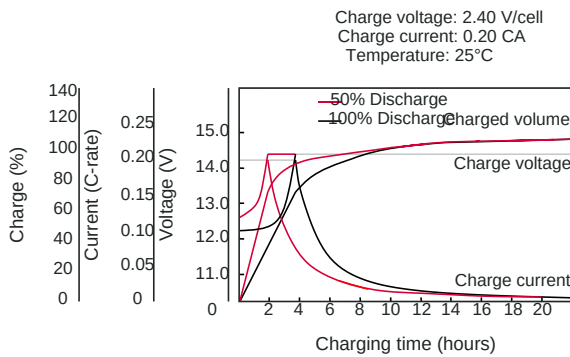
End voltage is per cell. Multiply W/cell by 6 for power per 12 V battery. Current is not divided by 6. Values are averages obtained within three charge/discharge cycles, not guaranteed minimums. Fully charge before testing. Product information may change without notice.

Characteristic curves

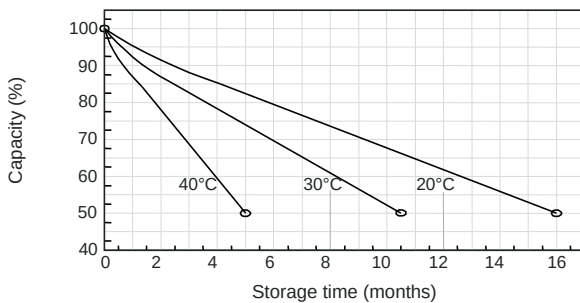
Discharge characteristics



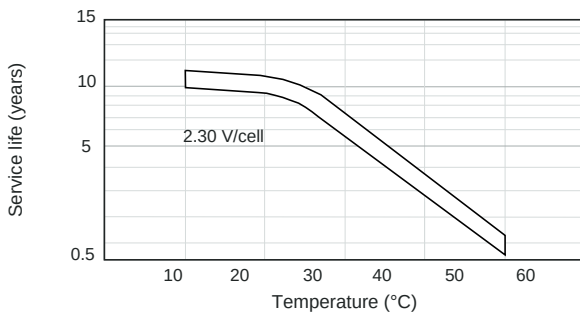
Charging characteristics



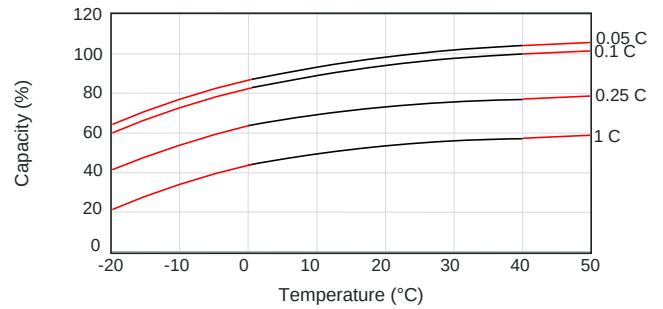
Self-discharge



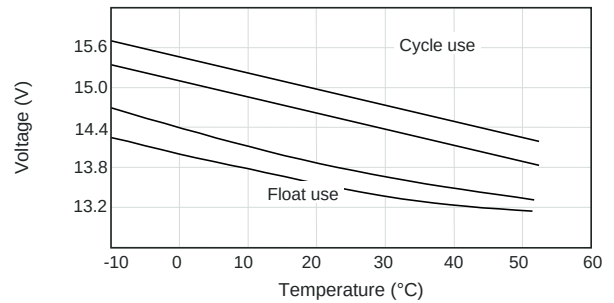
Float life vs. temperature



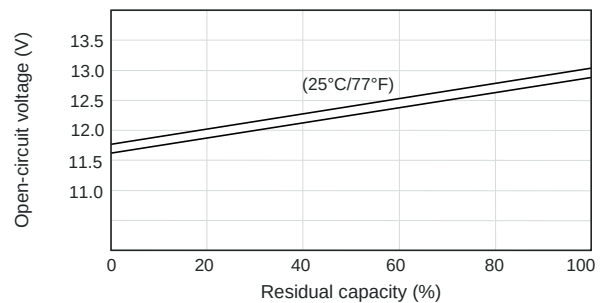
Capacity vs. temperature



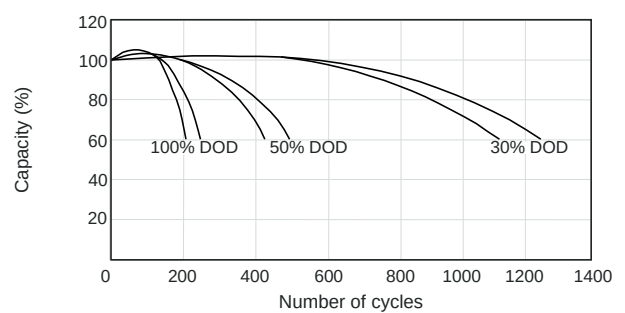
Charge voltage vs. temperature



Open-circuit voltage vs. capacity



Cycle life vs. depth of discharge



Typical characteristic curves. Model-specific ratings are given on pages 1 and 3. Design life is not a warranty.